

Unless stated otherwise, the Symbols have their usual meanings in context with the Subject. Assume suitably and state, additional data required, if any.
The Candidates, before starting to write the solutions, should please check the Question Paper for any discrepancy, and also ensure that they have been delivered the question paper of right course no. and right subject title.

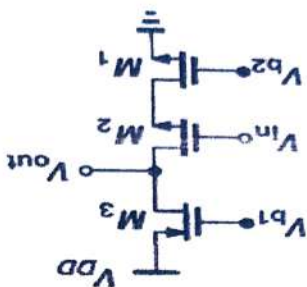
Note: Attempt all the questions with equal marks.

1.

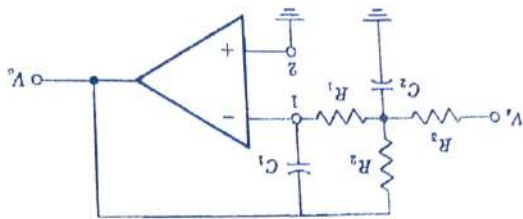
- i) Draw the equivalent circuit for the CE and CC configurations subject to the restrictions that $R_L = 0$. Show that the input impedance of the two circuits are identical.
- ii) Draw the circuits for the CE and CC configurations subject to the restriction that the input is open circuited. Show that the output impedances of the two circuits are identical.

OR

- 1a. Derive the expression of the voltage gain and output impedance of common source with source degeneration stage using small signal analysis.
- 1b. Assume all MOSFETs shown in the figure below are in saturation. Calculate the small signal voltage gain and output impedance of the circuit. (Consider CLM and Neglect Body bias)



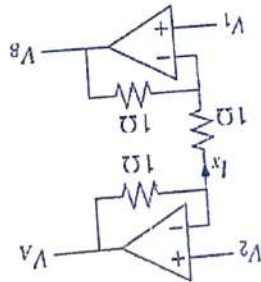
- 2a. Briefly explain the working of differential amplifier with active load (single ended output)
- 2b. For the circuit shown in the figure below, find the voltage gain V_o/V_s and input impedance as a function of R_s, b, R_e and R_L . Assume that $h_{oe}(R_e + R_L) \leq 0.1$



10

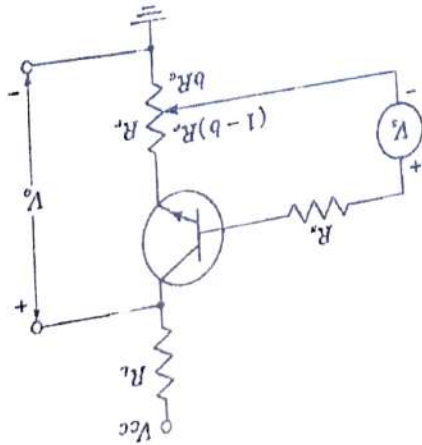
- ii) Using this circuit, design a second order Butterworth low-pass filter with $f_0 = 1$ KHz and low frequency voltage gain equal to -1
- i) Find the voltage gain $A_v = V_o / V_s$, the damping factor k and the cutoff frequency ω_c

4. The circuit shown in the figure below uses an ideal OPAMP
- 3b. An amplifier using an op-amp with a slew rate (SR) = 1.2 V/ μ S has a gain of 30 dB. If this amplifier has to faithfully amplify sinusoidal signals from dc to 25 KHz without introducing any slew-rate-induced distortion, then calculate the maximum value of the input signal level.



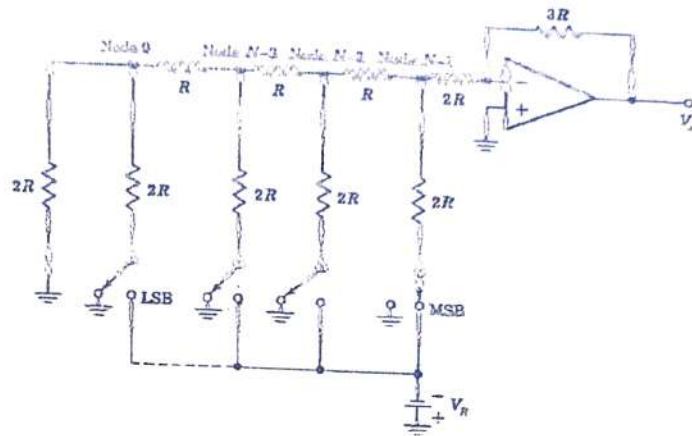
5

- 3a. Assume the op-amps in the circuit shown below are "ideal". Determine the current I_x if $V_2 = 2$ V and $V_1 = 1$ V. Also, determine the general expression of V_A and V_B in terms of V_1 and V_2 .



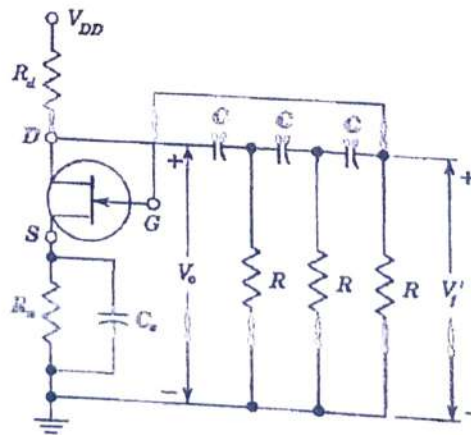
5

- 5a. Write a short note on compensation and stability in feedback amplifiers.
 5b. For the DAC converter shown in the figure below show that when the second most significant bit is 1 and all other bits are zero, the output is $V_o = V_R / 4$
- i) Find V_o if only the third MSB is 1
 ii) Find V_o if only the LSB is 1



OR

- 5a. Explain the practical utility of both ADC and DAC circuits. Also enumerate the different types of each along with detailed description of anyone.
 5b. Show that the feedback factor of the phase shift network shown in the figure below is $-\beta = \frac{V_f'}{V_o} = \frac{1}{1 - 5\alpha^2 - j(6\alpha - \alpha^3)}$ assuming that this network does not load the amplifier. Also, prove that the phase shift of $\frac{V_f'}{V_o}$ is 180° for $\alpha^2 = 6$ and that at this frequency $\beta = \frac{1}{29}$.



--- THE END ---